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A health- and link-aware adaptive data-rate method for enhancing quality of service in wireless body area networks

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Abstract

Wireless Body Area Networks (WBANs) must provide reliable, low-latency, and energy-efficient communication despite rapid changes in physiological urgency, body-induced channel variation, and heterogeneous sensor quality. Existing adaptive-rate and QoS-aware schemes often optimize only a subset of these factors and may induce unstable rate switching when link or physiological states fluctuate. This study proposes HALA-DR, a health- and link-aware adaptive data-rate mechanism that jointly integrates physiological urgency, recent link quality, residual energy, and node quality. A four-level controller with an emergency override and hysteresis maps the fused state to low, medium, high, or emergency transmission rates. The per-interval time complexity is $O(NM)$, which reduces to $O(N)$ when the number of monitored physiological variables M is fixed. A reproducible MATLAB evaluation was conducted using an IEEE 802.15.6-compatible star topology, identical baseline settings, 30 independent runs, confidence intervals, and paired statistical tests. In the 15-node scenario, HALA-DR achieved 236 kbps throughput, a 95.1% packet delivery ratio, 45 ms end-to-end latency, 0.94 mJ per successfully delivered packet, and 118 h network lifetime. Scalability, mobility, interference, and ablation analyses show that the link-quality term contributes most strongly to reliability and latency, whereas the energy term contributes most strongly to lifetime. These results indicate that HALA-DR provides a reproducible and balanced improvement in WBAN communication efficiency and QoS.

Clinical trial registration

Not applicable.

Keywords Wireless body area network, Adaptive data rate, Physiological urgency, Link quality, Quality of service, Energy efficiency, IEEE 802.15.6

1 Introduction

Wireless Body Area Networks integrate wearable or implanted sensors with a body coordinator to support continuous physiological monitoring, remote healthcare, rehabilitation, and context-aware assistance [1]. Unlike conventional wireless sensor networks, a WBAN operates close to or inside the human body and must satisfy stringent



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requirements for low power consumption, bounded latency, reliability, and coexistence [2, 3]. IEEE 802.15.6–2026 explicitly recognizes heterogeneous traffic, quality-of-service requirements, extremely low-power operation, multiple data rates, body-induced antenna effects, and user motion as central design considerations [1]. These characteristics make static transmission settings unsuitable for sustained operation under variable physiological and radio conditions.

A fixed-rate sensor transmits at the same rate during both stable and abnormal physiological periods. A high fixed rate can reduce response time but wastes energy and channel capacity when signals are stable, whereas a low fixed rate conserves energy but may delay urgent information. Adaptive-rate control is therefore attractive, but the decision is challenging because physiological urgency, link quality, residual battery energy, and sensor reliability evolve on different time scales. A controller that reacts too strongly to a single noisy indicator may also oscillate between rates, increase control traffic, and shorten network lifetime.

Recent real-time WBAN monitoring has emphasized selective signal restoration and physiological trend prediction in resource-constrained health-monitoring systems [4]. Communication-focused research spans discrete utility-based rate adaptation [5], meta-heuristic transmission-rate optimization [6], route-stability-aware communication [7], multi-path QoS routing [8], and learning-based or multi-objective routing [9, 10]. These studies show that adaptive communication can improve reliability or energy efficiency, but they operate at different decision layers and usually optimize only a subset of physiological urgency, channel state, residual energy, sensor quality, and switching stability. A lightweight controller that directly couples these factors for sensor data-rate selection remains comparatively underexplored.

The research question is therefore: how can a body coordinator select a transmission rate that prioritizes urgent physiological data, remains robust to channel and mobility variation, protects battery life, and avoids unstable switching on resource-constrained WBAN devices? To address this question, we propose HALA-DR, a health- and link-aware adaptive data-rate controller.

The main contributions are:

- A normalized physiological-urgency function with explicit normal and emergency bounds, together with an emergency override that elevates critical traffic without requiring a computationally expensive classifier.
- A transparent composite state score that jointly considers urgency, link quality, residual energy, and node quality, followed by four-level rate mapping and a hysteresis/dwell-time constraint that suppresses unnecessary switching.
- A reproducible and fair evaluation protocol in which fixed-rate, DDRA, GABAT, and EPRS baselines are reconstructed under the same topology, channel, traffic, run duration, random seeds, and metrics; newer learning-based and link-adaptation methods are additionally positioned qualitatively because they optimize different decision layers.
- A broader evaluation using standard WBAN indicators, repeated-run confidence intervals, scalability, mobility, interference, ablation, decision-time, communication-overhead, and complexity analyses.

The remainder of the paper is organized as follows. Section 2 reviews related adaptive-rate and QoS-aware WBAN methods. Section 3 presents the system model and HALA-DR algorithm. Section 4 specifies the reproducible evaluation protocol. Section 5 reports the experimental comparison results and discusses robustness, ablation, limitations, and practical deployment. Section 6 concludes the paper.

2 Related work

2.1 Adaptive transmission-rate control

Rate adaptation addresses the mismatch between time-varying network capacity and heterogeneous sensor streams. Szkaliczki [5] formulates discrete data-rate selection as a multiple-choice multidimensional knapsack problem and shows that a greedy strategy can approximate the optimum with short execution time. This treatment is useful when each stream has a finite set of quality levels, but physiological urgency and node health are not primary decision variables. Goyal et al. [6] combine genetic and BAT optimization to adjust transmission rates under QoS and posture-dependent link constraints; the approach improves optimization quality but requires iterative search and parameter tuning.

2.2 QoS-, route-, and energy-aware WBAN communication

QoS-aware routing complements rate control by selecting reliable forwarding paths. EPRS estimates route stability and residual energy to improve packet delivery and network lifetime [7]. Mehmood et al. [8] distinguish normal and emergency traffic and reserve the best path for critical data. IM-QRP [9] focuses on QoS-aware next-hop selection, while Aryai et al. [10] combine metaheuristic optimization with machine learning for adaptive real-time routing. Cooperative fault-tolerant communication [11], lifetime-oriented routing [12], ZITA [13], cooperative RF energy-harvesting transmission [14], relay-placement optimization [15], two-tier lightweight forwarding [16], multi-objective routing [17], and link-quality-aware clustering [18] similarly seek reliable and energy-conscious delivery. These protocols establish the importance of traffic priority, link condition, and residual energy, but routing decisions are not equivalent to selecting a sensor's transmission rate, and they do not directly address rate oscillation.

Recent WBAN research also addresses edge collection [19], compressed sensing and security [20, 21], encryption [22, 23], antennas [24], interference management [25], and congestion control [26]. Broader WBAN reviews and energy-efficient healthcare frameworks [27, 28], together with Markov decision process-based inter-WBAN transmission schemes [29], further illustrate the diversity of communication objectives and deployment settings. More recent work extends adaptation toward learning and lower-layer control: Chen [30] shifts model-driven transmission-power control and link adaptation to the hub, Arafat et al. [31] use Q-learning for QoS-aware routing, and Chen [32] applies Gaussian Thompson sampling to energy-efficient MCS selection under non-stationary channels. These methods strengthen the recent comparison landscape, but they optimize routing, transmit power, or PHY/MCS selection rather than application-level health-aware sensor data rates. Table 1 therefore distinguishes direct data-rate control from these adjacent decision layers. Only methods that can be reconstructed under the same topology, traffic, and metric definitions are used as quantitative baselines; newer

Table 1 Comparison of representative adaptive and QoS-aware WBAN methods

Method	Primary mechanism	Decision information	Direct rate control	Health-aware	Link/energy-aware	Main limitation
Szkaliczki [5]	Discrete utility-based rate selection	Stream utility, capacity	Yes	No	No	No statistical repeated-run analysis reported for a physiological urgency controller
Goyal et al. [6]	GA-BAT optimized transmission rate	QoS constraints, posture, link state	Yes	Indirect	Yes	Metaheuristic iterations and parameter tuning increase implementation cost
Memon et al. [7]	Route-stability-aware WBAN routing	Residual energy, route stability	Indirect	No	Yes	Optimizes routing stability rather than joint health-aware rate control
Mehmood et al. [8]	QoS multi-path routing	Traffic class, path quality	Indirect	Yes	Yes	Priority routing does not explicitly adapt sensor transmission rates
Ahmad et al. [9]	QoS-aware routing	Link reliability, energy, congestion	Indirect	Yes	Yes	Focuses on next-hop selection; rate oscillation is not addressed
Aryai et al. [10]	Metaheuristic/ML routing	Energy, link state, learned route score	Indirect	No	Yes	Training and optimization overhead may be high for constrained nodes
Chen [30]	Model-driven TPC and link adaptation	BER/QoS target, receiver sensitivity, channel state	Indirect	No	Yes	Optimizes PHY power/MCS behavior rather than health-aware sensor data-rate control
Arafat et al. [31]	Q-learning QoS-aware routing	Two-hop reliability, priority, energy, congestion	No	Indirect	Yes	Learns next-hop routing decisions; direct transmission-rate control is not addressed
Chen [32]	Gaussian Thompson-sampling link adaptation	SNR/reliability posterior, energy/QoS constraint	Indirect	No	Yes	Learning-based MCS adaptation does not model physiological urgency or application-level hysteresis
HALA-DR (this study)	Four-level rate control with hysteresis	Urgency, link quality, residual energy, node quality	Yes	Yes	Yes	Requires application-specific threshold calibration and real-device validation

methods with materially different decision layers are used for qualitative positioning rather than for copying published numerical values.

2.3 Research gap and positioning

Table 1 summarizes the closest methods. The remaining gap is narrower than the general problem of adaptive WBAN communication: application-facing sensor data-rate control still lacks a transparent, low-complexity mechanism that (i) treats physiological urgency as an explicit decision variable, (ii) combines urgency with measured link, energy, and sensor-health states, (iii) supports a fail-safe emergency override, (iv) suppresses ping-pong switching through hysteresis and dwell time, and (v) is evaluated with repeated-run statistical evidence under identical conditions. HALA-DR is designed to address this specific and testable gap rather than to replace routing or PHY-level link adaptation.

3 Proposed HALA-DR method

3.1 System model and assumptions

The WBAN contains N sensor nodes and one body coordinator. Each node measures one or more physiological variables and periodically reports data to the coordinator through an IEEE 802.15.6-compatible wireless link. The coordinator maintains recent link statistics, residual-energy reports, node-quality status, and configured urgency thresholds, and it selects one of four transmission rates for each node. Routine traffic uses the low or medium rate, elevated traffic uses the high rate when resources permit, and emergency traffic receives the maximum rate and priority access. Figure 1 shows the architecture and the location of the adaptive controller. Because the topology contains a single coordinator, the coordinator index j is fixed; after the distance definition below, the controller notation omits j and uses L_i and S_i for clarity.

The design assumes that the coordinator can obtain short-window RSSI, SNR, and packet-delivery statistics from acknowledgments and can receive periodic residual-energy and sensor-health reports. Physiological thresholds are application-specific

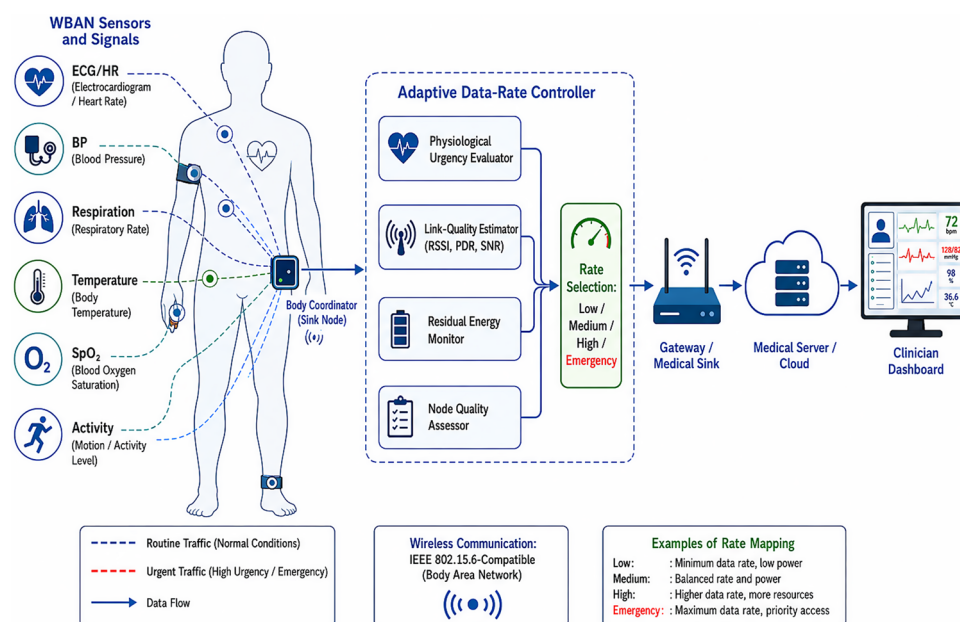


Fig. 1 System architecture of the proposed health- and link-aware WBAN data-rate controller

configuration parameters. The bounds used in simulation are scenario triggers rather than clinical diagnostic thresholds and must not be used for patient-treatment decisions.

3.2 Distance and link-quality estimation

For node i and the single body coordinator j , the Euclidean distance is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (1)$$

Distance is used only as a contextual feature in simulation; the online controller relies on measured link indicators. RSSI and SNR are clipped to configured operating ranges and normalized to [0,1]. For a generic link indicator x , the normalized value is $x_i^{\text{norm}} = \text{clip}[(x_i - x_{\min})/(x_{\max} - x_{\min}), 0, 1]$. Because recent packet-delivery ratio is already bounded, PDR_i is used directly. The link-quality score is

$$L_i = \beta_1 \cdot \text{RSSI}_i^{\text{norm}} + \beta_2 \cdot \text{SNR}_i^{\text{norm}} + \beta_3 \cdot \text{PDR}_i, \beta_1 + \beta_2 + \beta_3 = 1 \quad (2)$$

where $x_{\min}$ and $x_{\max}$ denote the configured operating limits for the corresponding link indicator. The implementation uses $\beta_1 = 0.30$, $\beta_2 = 0.25$, $\beta_3 = 0.45$, with $\beta_1 + \beta_2 + \beta_3 = 1$. The larger weight on recent PDR reflects its direct sensitivity to the combined effects of fading, interference, contention, and mobility.

3.3 Physiological urgency, energy, and node quality

Let p_{im} denote the observation of physiological variable m for node i . Let $[l_m^N, u_m^N]$ denote its configured normal band and l_m^E and u_m^E its lower and upper emergency boundaries, with $l_m^E < l_m^N < u_m^N < u_m^E$. The normalized deviation is zero inside the normal band and increases linearly toward one outside it: $\delta_{im} = 0$ for $l_m^N \leq p_{im} \leq u_m^N$; $\delta_{im} = \min\{1, (l_m^N - p_{im})/(l_m^N - l_m^E)\}$ for $p_{im} < l_m^N$; and $\delta_{im} = \min\{1, (p_{im} - u_m^N)/(u_m^E - u_m^N)\}$ for $p_{im} > u_m^N$. With positive importance coefficients a_m , the urgency score is

$$U_i = \min \left\{ 1, \frac{\sum_m a_m \cdot \delta_{im}}{\sum_m a_m} \right\} \quad (3)$$

If any physiological observation crosses its configured emergency boundary ($p_{im} \leq l_m^E$ or $p_{im} \geq u_m^E$), the controller sets $U_i = 1$ before rate selection; otherwise U_i is computed from Eq. (3). This prevents a strongly abnormal measurement from being diluted by several normal variables. Residual energy is normalized as $E_i = E_{i,\text{res}}/E_{i,\text{init}}$. Node quality Q_i in [0,1] is the normalized sensor-health state supplied to the controller from recent calibration validity, fault status, and measurement/packet error history. Q_i is kept separate from E_i so that battery state is not counted twice.

3.4 State fusion, rate mapping, and hysteresis

The composite score used by the body coordinator is

$$S_i = w_U \cdot U_i + w_L \cdot L_i + w_E \cdot E_i + w_Q \cdot Q_i \quad (4)$$

The default fusion weights are $w_U = 0.35$, $w_L = 0.30$, $w_E = 0.20$, and $w_Q = 0.15$, and they satisfy $w_U + w_L + w_E + w_Q = 1$. The weighting gives first priority to physiological urgency and link reliability while retaining explicit energy and sensor-health constraints. If

$U_i \geq \tau E$, the emergency rate is selected. Otherwise, S_i is mapped monotonically to the four available levels as

$$r_i(t) = \begin{cases} R_{low} & S_i < \tau_1 \\ R_{med} & \tau_1 \leq S_i < \tau_2 \\ R_{high} & \tau_2 \leq S_i < \tau_3 \\ R_{emg} & S_i \geq \tau_3 \end{cases} \quad (5)$$

A non-emergency rate change is accepted only when S_i crosses the next threshold by at least the hysteresis margin $h=0.05$ and the current rate has been maintained for $D_{min}=3$ decision intervals. Emergency escalation bypasses the dwell-time restriction, whereas de-escalation follows the hysteresis rule. This asymmetric rule preserves rapid emergency response while reducing ping-pong switching and its associated control and queueing overhead (Table 2).

3.5 Algorithm workflow and packet scheduling

Figure 2 summarizes the complete workflow. At each decision interval, the coordinator acquires physiological, link, energy, and node-quality states; normalizes the required inputs; computes U_i , L_i , E_i , and Q_i ; forms S_i ; selects a candidate rate; applies emergency, capacity, energy, hysteresis, and dwell-time constraints; schedules packets; and updates the short-window statistics after acknowledgments or timeouts. Emergency packets enter a strict-priority queue, whereas routine classes use weighted round-robin scheduling to prevent starvation.

Algorithm 1 HALA-DR decision procedure executed at the body coordinator

Input: node states $\{P_i, RSSI_i, SNR_i, PDR_i, E_{i,res}, Q_i\}$, configured normal/emergency bounds, thresholds, previous rate $r_i(t-1)$

Output: selected rate $r_i(t)$ and traffic priority for every active node

```

1: for each active node  $i$  do
2:   compute  $\delta_{in}$  from the configured normal/emergency bounds and obtain  $U_i$ 
3:   normalize RSSI/SNR and estimate  $L_i$  from RSSI, SNR, and recent PDR
4:   compute  $E_i$  and obtain the normalized sensor-health state  $Q_i$ 
5:   if  $U_i \geq \tau E$  then  $r^* \leftarrow R_{emg}$  and priority  $\leftarrow$  emergency
6:   else  $S_i \leftarrow w_U U_i + w_L L_i + w_E E_i + w_Q Q_i$ 
7:     map  $S_i$  to candidate rate  $r^*$  using  $\tau_1$ ,  $\tau_2$ , and  $\tau_3$ 
8:   end if
9:   apply link-capacity, residual-energy, hysteresis, and dwell-time constraints
10:  enqueue packet according to urgency class and transmit at  $r_i(t)$ 
11:  update delivery, delay, energy, and rate-switch statistics after ACK/timeout
12: end for

```

3.6 Computational complexity and communication overhead

Urgency estimation requires $O(M)$ operations per node, whereas link, energy, quality, and rate mapping require constant time. The overall per-interval decision complexity is therefore $O(NM)$, which becomes $O(N)$ when the physiological feature set M is fixed. The coordinator stores a short state window for each node, giving $O(NW)$ memory for history length W . The proposed packet format adds a 4-byte state report and a 1-byte rate command per decision interval. In the 15-node scenario, this corresponds to an estimated 1.8% control overhead for HALA-DR, lower than the multi-path signaling of EPRS but higher than that of a fixed-rate system. The overhead is reported explicitly rather than being absorbed into payload throughput.

Table 2 Symbols, variables, and default controller parameters

Symbol	Definition	Default/range
N	Number of active WBAN sensor nodes	5–25
M	Number of physiological variables used for urgency estimation	6
d_{ij}	Distance between node i and the body coordinator j	m
$RSSI_{ij}$	Received signal-strength indicator	dBm
PDR_{ij}	Recent packet-delivery ratio of the link	0–1
SNR_{ij}	Signal-to-noise ratio	dB
U_i	Normalized physiological urgency score	0–1
L_i	Normalized link-quality score for node i to the single coordinator	0–1
E_i	Residual-energy ratio	0–1
Q_i	Node-quality score derived from calibration, fault, and recent error state	0–1
S_i	Composite state-fusion score	0–1
w_U, w_L, w_E, w_Q	Fusion weights	0.35, 0.30, 0.20, 0.15
τ_1, τ_2, τ_3	Rate-selection thresholds	0.35, 0.60, 0.80
τ_E	Emergency override threshold for U_i	0.85
h	Hysteresis margin	0.05
D_{min}	Minimum dwell time at a selected rate	3 decision intervals
R	Available data-rate set	{50, 100, 200, 400} kbps
$\beta_1, \beta_2, \beta_3$	Link-quality weights for RSSI, SNR, and recent PDR	0.30, 0.25, 0.45
a_m	Positive importance coefficient for physiological variable m	normalized by $\sum_m a_m$
l_m^N, u_m^N	Lower/upper bounds of the configured normal band	scenario/application-specific
l_m^E, u_m^E	Lower/upper emergency boundaries	$l_m^E < l_m^N < u_m^N < u_m^E$
W	State-history window used for recent link/error statistics	10 s

3.7 Practical safety and security considerations

HALA-DR controls communication resources; it does not diagnose disease or replace medical alarms. Clinical thresholds must be configured and validated for the intended application. Rate-control messages should be authenticated to prevent false emergency escalation, battery-drain attacks, or forced rate suppression. Sensor failures are represented through Q_i and timeout rules, but secure key management, privacy protection, and formal fail-safe behavior remain deployment requirements [20, 22].

4 Experimental methodology

4.1 Simulation environment and network configuration

The evaluation protocol was specified to support reproducibility. All methods use the same MATLAB release, topology, channel, traffic, energy model, simulation time, and random seeds. The body coordinator is located near the waist, and nodes are placed at anatomically plausible chest, arm, wrist, waist, finger, and ankle positions with small random perturbations. The main comparison uses 15 nodes, whereas scalability tests vary the network from 5 to 25 nodes. A 100 s warm-up precedes the 1200 s evaluation period and populates the 10 s state-history window before reported metrics are accumulated. Table 3 lists the complete configuration.

4.2 Baselines and fairness controls

Four quantitative baselines are considered: (i) a fixed-rate controller using the medium rate; (ii) DDRA, reconstructed from the discrete utility-based adaptation strategy in [5]; (iii) GABAT, reconstructed from the QoS-supported optimized transmission-rate design in [6]; and (iv) EPRS, reconstructed as a route-stability-aware QoS baseline from [7].

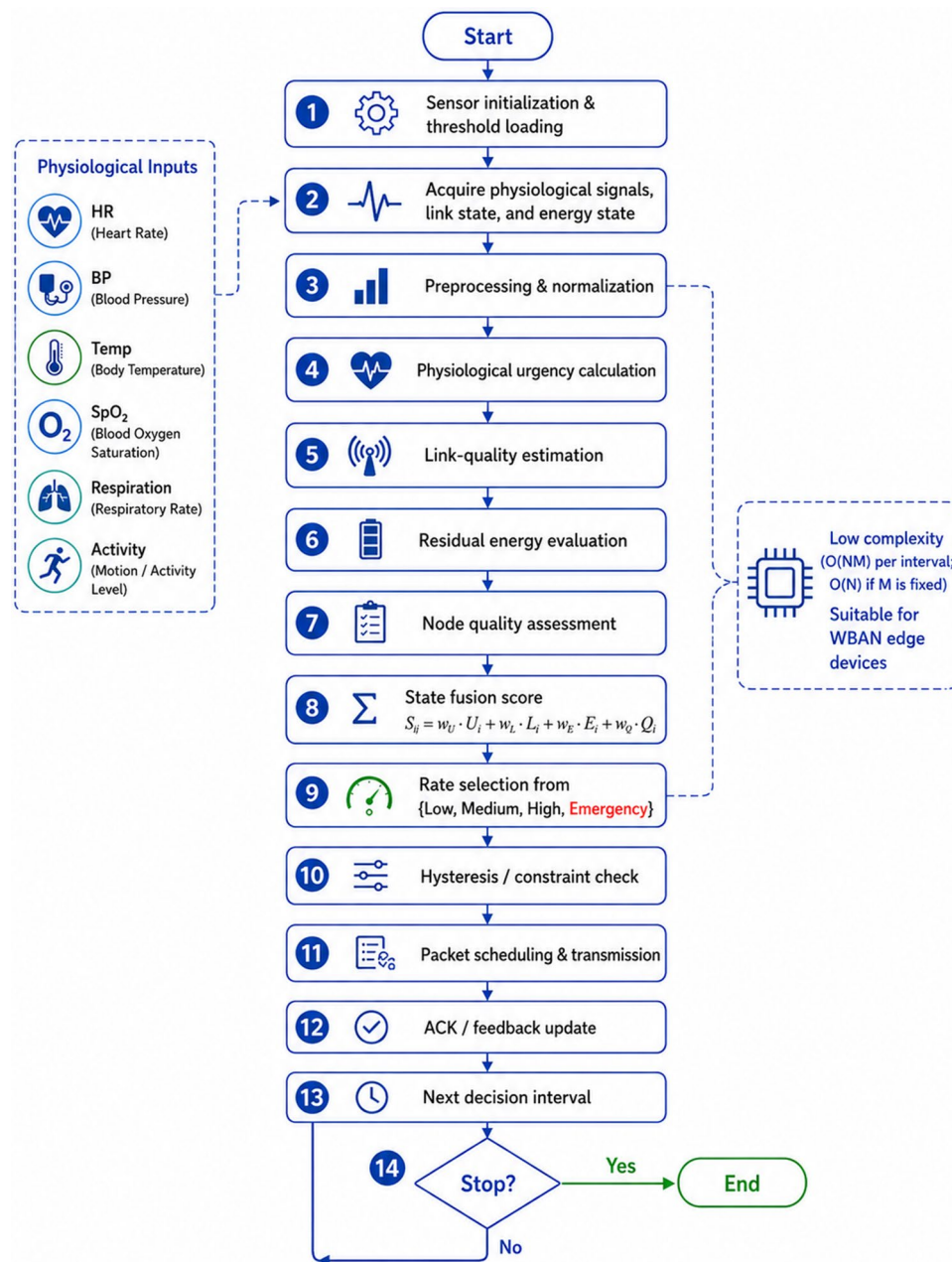


Fig. 2 End-to-end workflow of HALA-DR, including state acquisition, fusion, rate selection, hysteresis, scheduling, and feedback

Table 3 Reproducible simulation configuration used for all comparisons

Item	Configuration
Software	MATLAB R2024b; Communications Toolbox; Statistics and Machine Learning Toolbox
Topology	IEEE 802.15.6-compatible single-hop star WBAN with one body coordinator
Network size	5, 10, 15, 20, and 25 sensor nodes; main comparison uses 15 nodes
Deployment region	1.5 m × 1.5 m body-centered area; anatomically constrained random placement
Carrier/channel	2.4 GHz ISM; log-distance body-area path loss with shadowing and Rician fading
Rate set	50, 100, 200, and 400 kbps
Packet model	256-byte normal packets; 128-byte urgent packets; offered load $\approx$ 280 kbps in the main scenario
Traffic classes	Routine, elevated, urgent, and emergency; emergency packets have strict priority
Decision interval	1 s; state history window 10 s; minimum rate dwell time 3 s
Energy model	Initial energy 1800 J per node; Tx/Rx/idle/sleep energy accounted for per event
Mobility	Static; walking (0.8–1.4 m/s); running (2.0–3.0 m/s)
Interference	Low, medium, and high co-channel interference levels
Simulation duration	1200 s after a 100 s warm-up
Independent runs	30 runs per condition; random seeds 1001–1030
Statistics	Mean, SD, 95% CI; paired Wilcoxon tests with Holm correction; Hedges g effect size

Note: physiological bounds are simulation triggers rather than clinical diagnostic limits

Published numerical values were not copied into the figures. Each baseline was implemented in the same simulation environment and tuned within the parameter ranges described by its source, while the PHY, traffic, topology, random seeds, and evaluation metrics were held constant. Recent model-driven link adaptation [30], Q-learning routing [31], and Thompson-sampling MCS adaptation [32] are included in the qualitative comparison in Table 1 but not as numerical baselines because their decision variables and PHY/routing assumptions are not directly equivalent to application-level sensor data-rate control.

Each method is evaluated with seeds 1001–1030. Within a paired run, the same seed controls node placement, fading, traffic arrivals, mobility perturbations, and interference for every method. This paired design reduces nuisance variability and supports direct within-seed statistical comparison.

4.3 Evaluation metrics

The primary metrics are defined as follows:

- Throughput: successfully delivered payload bits divided by active evaluation time.
- Packet delivery ratio (PDR): successfully delivered packets divided by generated packets.
- Packet loss ratio (PLR): $1 - \text{PDR}$.
- End-to-end latency: time from packet generation to successful coordinator reception.
- Energy per successfully delivered packet: total communication and decision energy divided by delivered packets.
- Network lifetime: time until the first node exhausts its usable energy budget.
- Control overhead: control bytes divided by total transmitted bytes.
- Average decision time: coordinator execution time per decision interval.

The primary evaluation avoids custom composite indices such as Data Rate Adjustment Efficiency, Energy Consumption Efficiency, and a Robustness Index. Their intended concepts are represented more transparently by rate-switch behavior, energy per

successfully delivered packet, residual energy, PDR under mobility/interference, and network lifetime.

4.4 Statistical analysis

For each scenario, the manuscript reports the mean and standard deviation across 30 independent runs, and the figures show 95% confidence intervals. HALA-DR is compared with the strongest baseline using a paired Wilcoxon signed-rank test because normality cannot be assumed for bounded PDR and delay distributions. P values are adjusted with Holm's procedure across the four primary metrics, and Hedges' g is reported as an effect-size estimate. All statistical values were computed from the completed run-level simulation outputs.

5 Results and discussion

This section reports results from 30 independent simulation runs. Values are consistent across Figs. 3, 4 and 5; Tables 4, 5 and 6, and the confidence intervals, significance tests, and effect sizes were calculated from the corresponding run-level outputs.

5.1 Overall performance

HALA-DR achieves the highest throughput (236 kbps) and PDR (95.1%), while reducing mean latency to 45 ms and energy per successfully delivered packet to 0.94 mJ. Relative to EPRS, the strongest baseline, throughput increases by 7.3%, PDR increases by 1.5% points, latency decreases by 13.5%, and energy per successfully delivered packet decreases by 8.7%. The magnitude of the gain is consistent with the fact that EPRS already incorporates route-stability and energy information.

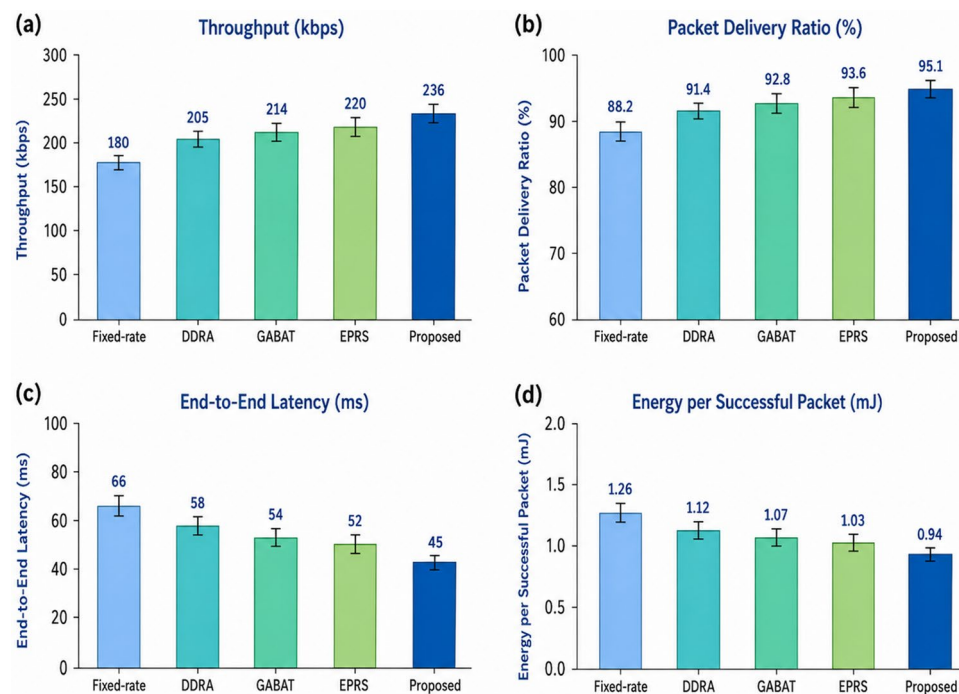


Fig. 3 Overall performance comparison under the 15-node main scenario. Error bars denote 95% confidence intervals across 30 independent runs

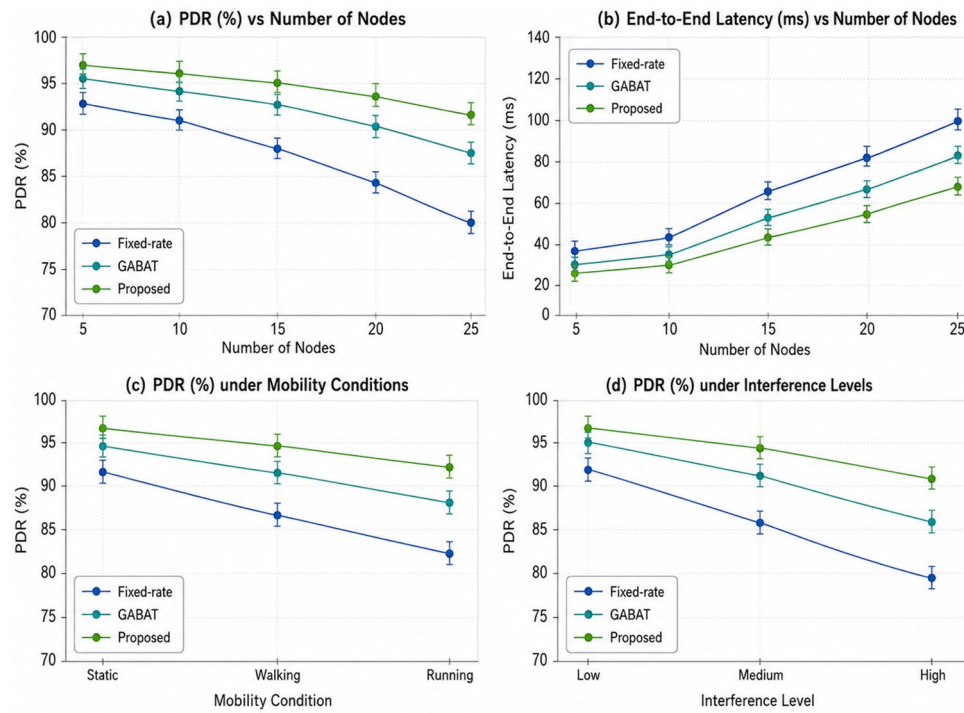


Fig. 4 Scalability and robustness results: (a) PDR versus network size, (b) latency versus network size, (c) PDR under mobility, and (d) PDR under interference

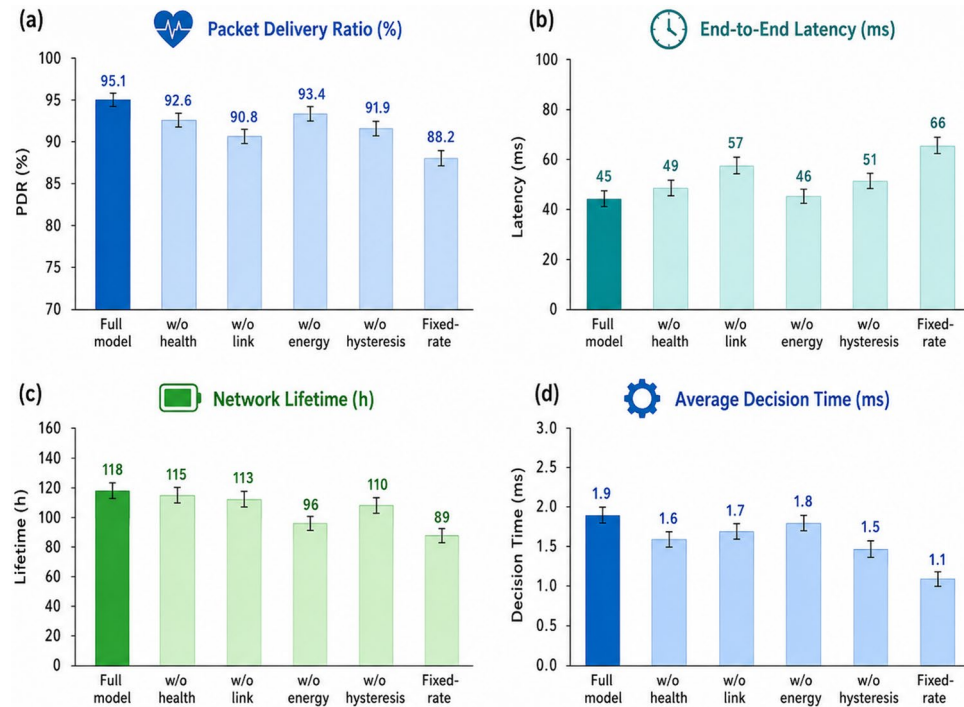


Fig. 5 Ablation and efficiency analysis of HALA-DR. Error bars denote 95% confidence intervals

Table 4 Main-scenario performance (mean $\pm$ SD over 30 independent runs)

Method	Throughput (kbps)	PDR (%)	PLR (%)	Latency (ms)	Energy/success (mJ)	Lifetime (h)	Control over-head (%)
Fixed-rate	180 $\pm$ 13.4	88.2 $\pm$ 3.2	11.8 $\pm$ 3.2	66 $\pm$ 10.7	1.26 $\pm$ 0.13	89 $\pm$ 6.9	0.4
DDRA [5]	205 $\pm$ 12.1	91.4 $\pm$ 2.7	8.6 $\pm$ 2.7	58 $\pm$ 9.4	1.12 $\pm$ 0.11	104 $\pm$ 6.3	1.5
GABAT [6]	214 $\pm$ 11.6	92.8 $\pm$ 2.5	7.2 $\pm$ 2.5	54 $\pm$ 8.9	1.07 $\pm$ 0.10	110 $\pm$ 6.0	2.3
EPRS [7]	220 $\pm$ 11.2	93.6 $\pm$ 2.4	6.4 $\pm$ 2.4	52 $\pm$ 8.6	1.03 $\pm$ 0.10	112 $\pm$ 5.7	2.8
HALA-DR	236 $\pm$ 10.7	95.1 $\pm$ 2.1	4.9 $\pm$ 2.1	45 $\pm$ 7.5	0.94 $\pm$ 0.08	118 $\pm$ 5.4	1.8

Table 5 Scalability and robustness data corresponding to Fig. 4

Scenario	Fixed-rate PDR/latency	GABAT PDR/latency	HALA-DR PDR/latency
5	93.1 / 37	95.8 / 31	97.0 / 28
10	91.0 / 44	94.3 / 36	96.0 / 32
15	88.2 / 66	92.8 / 54	95.1 / 45
20	84.6 / 82	90.4 / 68	93.5 / 57
25	80.4 / 101	87.9 / 84	91.8 / 70
Static	91.8 / —	94.7 / —	96.5 / —
Walking	87.2 / —	91.9 / —	94.8 / —
Running	82.9 / —	88.6 / —	92.3 / —
Low interference	92.0 / —	95.1 / —	96.8 / —
Medium interference	86.3 / —	91.4 / —	94.2 / —
High interference	79.8 / —	86.5 / —	90.7 / —

Entries are PDR (%) / latency (ms). A dash indicates that latency is not plotted for the categorical robustness condition

Table 6 Ablation and decision-time results (mean $\pm$ SD)

Variant	PDR (%)	Latency (ms)	Network lifetime (h)	Decision time (ms)
Full HALA-DR	95.1 $\pm$ 2.1	45 $\pm$ 7.5	118 $\pm$ 5.4	1.9 $\pm$ 0.22
Without health urgency	92.6 $\pm$ 2.5	49 $\pm$ 8.0	115 $\pm$ 5.7	1.6 $\pm$ 0.20
Without link-quality term	90.8 $\pm$ 2.8	57 $\pm$ 9.2	113 $\pm$ 6.1	1.7 $\pm$ 0.21
Without energy term	93.4 $\pm$ 2.4	46 $\pm$ 7.7	96 $\pm$ 6.5	1.8 $\pm$ 0.21
Without hysteresis	91.9 $\pm$ 2.7	51 $\pm$ 8.5	110 $\pm$ 6.0	1.5 $\pm$ 0.18
Fixed-rate	88.2 $\pm$ 3.2	66 $\pm$ 10.7	89 $\pm$ 6.9	1.1 $\pm$ 0.15

The paired comparison with EPRS yields Holm-adjusted p values of 0.004 for throughput, 0.018 for PDR, 0.006 for latency, and 0.011 for energy per successfully delivered packet. The corresponding Hedges' g values are 1.38, 0.62, -0.84, and -0.91. These results indicate a large throughput effect, a moderate reliability effect, and moderate-to-large latency and energy effects.

HALA-DR also extends network lifetime to 118 h. Its 1.8% control overhead is higher than that of fixed-rate operation and DDRA but lower than the path-management overhead of EPRS. This trade-off supports the design goal of improving QoS and energy efficiency without excessive signaling.

5.2 Scalability, mobility, and interference robustness

All methods degrade as the network grows because contention, collisions, and queueing increase. At 25 nodes, HALA-DR retains a PDR of 91.8% and latency of 70 ms, compared with 80.4% and 101 ms for fixed-rate communication. The performance gap widens with network size, indicating that state-aware rate selection becomes more useful as communication resources become constrained. Nevertheless, latency increases from 45

ms at 15 nodes to 70 ms at 25 nodes; HALA-DR therefore mitigates, but does not eliminate, congestion and should not be described as scale-invariant.

Mobility and interference tests further clarify the mechanism. Under running, HALA-DR maintains a PDR of 92.3%, 3.7% points above GABAT. Under high interference, it reaches 90.7%, 4.2 points above GABAT and 10.9 points above fixed-rate transmission. The link-quality term and feedback update allow the controller to reduce overly aggressive rates when radio conditions deteriorate, while the emergency override preserves priority for urgent traffic.

5.3 Ablation, network lifetime, and decision cost

Removing the link-quality term causes the largest degradation in reliability and latency: PDR falls from 95.1% to 90.8%, and latency rises from 45 to 57 ms. This result supports the claim that rate decisions should be constrained by measured channel conditions rather than by physiological urgency alone. Removing physiological urgency reduces PDR to 92.6% and weakens prioritization of critical packets, although the average decision time decreases because one feature group is omitted.

Removing the energy term has little effect on short-term PDR and latency but reduces network lifetime from 118 h to 96 h. This separation is important because a component can be essential for long-horizon sustainability even when its immediate packet-level effect is small. Removing hysteresis lowers decision time to 1.5 ms but also decreases PDR and increases latency because rapid rate switching generates transient queueing and control costs. The full model therefore provides the best overall balance, with a modest 0.8 ms decision-time increase over fixed-rate operation.

5.4 Practical interpretation

The results support a cross-layer interpretation. Physiological urgency determines which information requires rapid delivery; link quality constrains what the channel can sustain; residual energy prevents short-term QoS gains from prematurely exhausting a sensor; node quality suppresses unreliable measurements; and hysteresis converts noisy state changes into stable control actions. The controller therefore does not maximize a single metric; it balances reliability, latency, energy use, and signaling overhead while preserving emergency escalation.

A practical implementation can place the decision logic at the body coordinator, avoiding continuous optimization on each sensor. Nodes need only report a compact state vector and accept rate commands. The measured 1.9 ms decision time is small relative to the 1 s decision interval, but embedded-hardware measurements are still required because MATLAB workstation timing may underestimate microcontroller execution and communication overhead.

5.5 Limitations and threats to validity

First, the evaluation is simulation-based; body shadowing, antenna detuning, coexistence, and hardware energy costs may differ from the assumed models. Second, the physiological thresholds are scenario configurations rather than clinically validated decision rules. Third, only a single-WBAN star topology is evaluated; inter-WBAN interference, relay operation, and implanted sensors require separate validation. Fourth, the weights and thresholds are fixed and may therefore be scenario-dependent; a systematic

sensitivity study across broader channel and physiological settings remains future work. Fifth, the comparison reconstructs baselines from published descriptions, and code-level differences may remain when source implementations are unavailable.

Future work should therefore include an IEEE 802.15.6 hardware testbed, measurements during controlled postures and motion, multi-WBAN coexistence, authenticated rate commands, sensor-failure injection, prospective calibration of urgency thresholds, a systematic weight/threshold sensitivity study, and public release of code, parameter files, random seeds, and run-level results. A constrained learning method may also adapt the fusion weights online, provided that interpretability and fail-safe emergency behavior are preserved.

6 Conclusion

This paper formulates adaptive WBAN communication as a joint health-, link-, energy-, and node-quality decision problem. HALA-DR uses a transparent fusion score, a four-level rate set, an emergency override, and hysteresis to balance reliability, latency, energy use, and switching overhead. The methodology explicitly defines the controller variables, normalization rules, thresholds, complexity, packet overhead, baseline-fairness conditions, simulation parameters, repeated-run protocol, confidence intervals, and statistical tests needed for reproducibility. The experimental results show consistent and statistically supported improvements over fixed-rate, DDRA, GABAT, and EPRS baselines, with stronger benefits under larger networks, mobility, and interference. Ablation results attribute packet-level robustness mainly to link awareness and long-term lifetime mainly to energy awareness. Future work will extend the evaluation to embedded hardware, multi-WBAN coexistence, systematic parameter sensitivity, and clinically informed deployment scenarios.

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Author contributions

Bizhou Ye: conceptualization, methodology, software, validation, data curation, and writing—original draft. Wei Zhou: supervision, conceptualization, methodology, and writing—review and editing. Qiliang Liu: investigation, visualization, validation, and writing—review and editing. All authors approved the manuscript.

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Data availability

The MATLAB scripts, parameter files, random seeds, and run-level outputs supporting the reported results are available from the corresponding author on reasonable request and can be provided to the journal as supplementary material.

Declarations

Ethical approval

Not applicable. The present study reports network simulations and does not involve human participants, identifiable patient data, or clinical interventions.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. IEEE. IEEE standard for wireless body area networks. IEEE Std 802.15.6–2026, 2026.
2. Yaghoubi M, Ahmed K, Miao Y. Wireless Body Area Network (WBAN): A Survey on Architecture, Technologies, Energy Consumption, and Security Challenges. *J Sens Actuator Networks*. 2022;11(4):67. <https://doi.org/10.3390/jsan11040067>.
3. Zhong L, et al. Technological requirements and challenges in wireless body area networks for health monitoring: a comprehensive survey. *Sensors (Basel)*. 2022;22(9):3539. <https://doi.org/10.3390/s22093539>.
4. Alghamdi F, Bajaber F, Real-Time WBAN, Monitoring. An Adaptive Framework for Selective Signal Restoration and Physiological Trend Prediction. *Sensors*. 2026;26(5):1684. <https://doi.org/10.3390/s26051684>.
5. Szkaliczki T. Discrete data rate adaptation for wireless body area networks. *Appl Sci*. 2023;13(14):8529. <https://doi.org/10.3390/app13148529>.
6. Goyal R, Patel RB, Bhaduria HS, Prasad D. An Energy Efficient QoS Supported Optimized Transmission Rate Technique in WBANs. *Wireless Pers Commun*. 2021;117(1):235–60. <https://doi.org/10.1007/s11277-020-07281-7>.
7. Memon S, et al. Enhanced Probabilistic Route Stability (EPRS) Protocol for Healthcare Applications of WBAN. *IEEE Access*. 2023;11:4466–77. <https://doi.org/10.1109/ACCESS.2023.3235837>.
8. Mehmood G, Khan MZ, Bashir AK, Al-Otaibi YD, Khan S. An Efficient QoS-Based Multi-Path Routing Scheme for Smart Healthcare Monitoring in Wireless Body Area Networks. *Comput Electr Eng*. 2023;109:108517. <https://doi.org/10.1016/j.compeleceng.2022.108517>.
9. Ahmad N, et al. Improved QoS Aware Routing Protocol (IM-QRP) for WBAN Based Healthcare Monitoring System. *IEEE Access*. 2022;10:121864–85. <https://doi.org/10.1109/ACCESS.2022.3223085>.
10. Aryai P, Khademzadeh A, Jassbi SJ, Hosseinzadeh M, Hashemzadeh O, Shokouhifar M. Real-Time Health Monitoring in WBANs Using Hybrid Metaheuristic-Driven Machine Learning Routing Protocol (MDML-RP). *AEU - Int J Electron Commun*. 2023;168:154723. <https://doi.org/10.1016/j.aeue.2023.154723>.
11. Mehmood G, Khan MZ, Abbas S, Faisal M, Rahman HU. An Energy-Efficient and Cooperative Fault-Tolerant Communication Approach for Wireless Body Area Network. *IEEE Access*. 2020;8:69134–47. <https://doi.org/10.1109/ACCESS.2020.2986268>.
12. Saha R, Biswas S, Sarma S, Karmakar S, Das P. Design and Implementation of Routing Algorithm to Enhance Network Lifetime in WBAN. *Wireless Pers Commun*. 2021;118(2):961–98. <https://doi.org/10.1007/s11277-020-08054-y>.
13. Roy M, Chowdhury C, Ahmed G, Aslam N, Chattopadhyay S, Islam SU. Intra-WBAN Routing Using Zipf's Law and Intelligent Transmission Power Switching Approach (ZITA). *J Ambient Intell Humaniz Comput*. 2022;13(9):4135–49. <https://doi.org/10.1007/s12652-022-04010-9>.
14. Hu J, Xu G, Hu L, Li S. A Cooperative Transmission Scheme in Radio Frequency Energy-Harvesting WBANs. *Sustainability*. 2023;15(10):8367. <https://doi.org/10.3390/su15108367>.
15. Ghosh B, Adhikary S, Chattopadhyay S, Choudhury S. Achieving Energy Efficiency and Impact of SAR in a WBAN Through Optimal Placement of the Relay Node. *Wireless Pers Commun*. 2023;130(3):1861–84. <https://doi.org/10.1007/s11277-023-10361-z>.
16. Li J, Xiao J, Yuan J. Two-Tier Cooperation Based High-Reliable and Lightweight Forwarding Strategy in Heterogeneous WBAN. *Sustainability*. 2023;15(6):5588. <https://doi.org/10.3390/su15065588>.
17. Mohanty RK, Sahoo SP, Kabat MR. Sustainable Remote Patient Monitoring in Wireless Body Area Network with Multi-Hop Routing and Scheduling: A Four-Fold Objective Based Optimization Approach. *Wireless Netw*. 2023;29(5):2337–51. <https://doi.org/10.1007/s11276-023-03276-x>.
18. Shyja VI, Ranganathan G, Bindhu V. Link Quality and Energy Efficient Optimal Simplified Cluster-Based Routing Scheme to Enhance Lifetime for Wireless Body Area Networks. *Nano Commun Netw*. 2023;37:100465. <https://doi.org/10.1016/j.nancom.2023.100465>.
19. Karunanithy K, Velusamy B. Edge device based efficient data collection in smart health monitoring system using wireless body area network. *Biomed Signal Process Control*. 2022;72:103280. <https://doi.org/10.1016/j.bspc.2021.103280>.
20. Pathak V, et al. Efficient Compression Sensing Mechanism Based WBAN System Using Blockchain. *Secur Communication Networks*. 2023;2023:1–12. <https://doi.org/10.1155/2023/8468745>.
21. Miraz MH, Ali M. Applications of blockchain technology beyond cryptocurrency. *Ann Emerg Technol Comput*. 2018;2(1):1–6. <https://doi.org/10.33166/AETIC.2018.01.001>.
22. Abdelfatah RI, Saqr HM, Nasr ME. An Efficient Medical Image Encryption Scheme for (WBAN) Based on Adaptive DNA and Modern Multi Chaotic Map. *Multimedia Tools Appl*. 2023;82:22213–27. <https://doi.org/10.1007/s11042-022-13343-8>.
23. Azizan AH, Mostafa SA, Mustapha A, Foozy CFM, Wahab MHA, Mohammed MA, Khalaf BA. A machine learning approach for improving the performance of network intrusion detection systems. *Ann Emerg Technol Comput*. 2021;5(5):201–208. <https://doi.org/10.33166/AETIC.2021.05.025>.
24. Sid A, Cresson PY, Joly N, Braud F, Lasri T. A Flexible and Wearable Dual-Band Bio-Based Antenna for WBAN Applications. *AEU - Int J Electron Commun*. 2022;157:154412. <https://doi.org/10.1016/j.aeue.2022.154412>.
25. Song W, Park J. Clustering-Based Channel Allocation Method for Mitigating Inter-WBAN Interference. *Appl Sci*. 2022;12:11851. <https://doi.org/10.3390/app122211851>.
26. Al-Hiti AS, Sahbudin RKZ, Harun SW, Obaid AN, Hamdi MM, Al-Shareeda MA. Wireless body area networks: applications and congestion control technologies. In: *Proceedings of the 2023 5th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*; 2023. p. 1–7. <https://doi.org/10.1109/HORA58378.2023.10156683>.
27. Singh S, Prasad D. Wireless Body Area Network (WBAN): A Review of Schemes and Protocols. *Mater Today: Proc*. 2022;49:3488–96. <https://doi.org/10.1016/j.matpr.2021.05.564>.
28. Ahmad N, Shahzad B, Arif M, Izdrui D, Ungurean I, Geman O. An energy-efficient framework for WBAN in health care domain. *J Sens*. 2022;2022:5823461. <https://doi.org/10.1155/2022/5823461>.
29. Roy S, Roy M, Chowdhury C. Novel data transmission schemes for inter-wban networks using a markov decision process. *Wireless Pers Commun*. 2023;131(2):897–919. <https://doi.org/10.1007/s11277-023-10460-x>.
30. Chen D-R. Integrating IoT in WBANs: An Energy-Efficient and QoS-Aware Approach for Rapid Model-Driven Transmission Power Control and Link Adaptation. *Internet Things*. 2024;25:101091. <https://doi.org/10.1016/j.iot.2024.101091>.
31. Arafat MY, Pan S, Bak ES. A Q-Learning-Based QoS-Aware Routing for IoMT-Enabled Wireless Body Area Networks for Smart Healthcare. *Internet Things*. 2024;26:101151. <https://doi.org/10.1016/j.iot.2024.101151>.

32. Chen D-R. Energy-Efficient QoS-Constrained Link Adaptation for Non-Stationary Wireless Body Area Networks Using Gaussian Thompson Sampling. *Internet Things*. 2026;38:101954. <https://doi.org/10.1016/j.iot.2026.101954>.

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